Appendix A Reproducibility and evidence sources

The original research profile is fixed at revision `09225c95c676ab3cae1c4b9586d74f7f803f519f` by `data_manifest_v1.json`. The post-review profile at `6e4378bef9a98037c20d1381eb9c7b61462a6578` uses `data_manifest_post_review_v1.json` for revised grading, controller validation and behavior checks. Both preserve the collected corpus, fitted predictor and recorded generation outcomes. Each live manifest binds the sources actually executed in its `locked_code` directory.

Table 15. Claim evidence sources.

Claim or output	Authoritative repository source
Tournament corpus	data_manifest_v1.json and ultimate_tournament_manifest.json
Boundary and matched effects	research/outputs/thesis_v1/evidence/*.csv
Proofs and exact finite checks	research/mathematical_foundations.md; test_mathematical_foundations.py
Controller execution	online_stopping_controller.py; online_generation.py
Latency and replay	research/outputs/semester2/online_stopping_20261002/
Adversarial bank	research/adversarial_tasks_v1.jsonl; adversarial_gold_v1.jsonl
Archived policy failures	research/reports/thesis_failure_audit_v1/audit_summary.json
Deployable prefix predictor	research/outputs/semester2/prefix_model_v1/
PDF authoring	tools/build_master_thesis.py; tools/render_thesis_math.mjs

The electronic supplement, `ThesisDocs/formal/supplements/v5/reproduction_and_history.txt`, gives commands for

detached checkouts, data verification and isolated reanalysis. Verification uses `tools/freeze_research_data.py`; `tools/recompute_thesis_evidence.py` reconstructs tables, and `tools/verify_post_review_behavior.py` checks saved controller behavior. Analysis outputs go to new scratch directories, and uncertainty analysis operates on copied live panels.

The recorded receipts distinguish thirty historical grader cases from ninety-eight revised tests, and 107 passing historical-suite tests from 243 post-review repository tests. Saved-prefix verification reproduces 480 decision histories across four live collections; reused baseline rows are not additional generations. These counts identify their source versions, rather than measuring corpus-wide label validity.

The fifty-two standardized trace files are mandatory freeze inputs. Supporting evidence includes the variable-horizon matrix, paired study arms, stored predictions and live event ledgers. Exact-byte and canonical-LF hashes identify selected evidence; they do not establish bit-identical regeneration of the partially recorded historical GPU environment. The current authoring manifests separately identify the rendering environment and exact PDF bytes.

Appendix B Claim scope and mathematical review

Table 16. Claim scope and required evidence.

Claim	Evidence required	Supported interpretation
Binary repair-corruption identity	Common transition panel or conditional probability proof	Exact decomposition in Chapter 2
General optimal stopping	Full conditional continuation law and all costs	Finite-horizon theorem; not a calibrated learned deployment
First drift crossing is optimal	Pathwise persistence after the crossing	Conditional theorem; not established universally by the empirical curves
0.955156 AUC	Stored historical tournament result	Retrospective non-nested diagnostic
Causal detector ranking	Task-grouped causal sequence outputs	Internal development evaluation
Replay completion-token saving	Frozen traces and selected prefixes	Counterfactual development quantity
Live completion-token saving	Actual generation events from both arms	Model-, task-, and protocol-specific measurement
No meaningful accuracy loss	Prespecified tolerance and adequate paired confidence interval	Not inferred merely from observed equality
Grader regression coverage	Versioned test inputs and execution receipts	Thirty historical cases and ninety-eight revised tests cover specified parser behaviors; corpus-wide validity requires separate adjudication

Appendix C Paired accuracy uncertainty

For iid task pairs, let I indicate an incorrect baseline answer paired with a correct active answer, and W a correct baseline answer paired with an incorrect active answer. The accuracy difference is $\delta = \pi_I - \pi_W$, where $\pi_I = \Pr(I = 1)$ and $\pi_W = \Pr(W = 1)$. The indicators are mutually exclusive within a task; their independence is not assumed.

Across n independent pairs, each marginal discordance count is binomial. Exact two-sided 97.5% Clopper-Pearson intervals $[L_I, U_I]$ for π_I and $[L_W, U_W]$ for π_W each have noncoverage probability at most 0.025 [15]. A union bound gives simultaneous coverage at least 0.95 regardless of dependence between the counts. Subtraction yields

$$\delta \in [L_I - U_W,\ U_I - L_W].$$

This is the conservative 95% paired interval in the live tables. Exact multinomial enumeration supplements the binomial coverage argument. With no discordances, both lower bounds are zero and both upper bounds equal $1 - 0.0125^{1/n}$. Observed equality therefore leaves nonzero uncertainty; no noninferiority tolerance or hypothesis was prespecified.

The twenty handpicked traps form a fixed challenge bank, so the iid-reference interval gives no randomized adversarial-population coverage. Completion-token intervals bootstrap paired tasks and the ratio of total token differences to baseline tokens; these are descriptive resampling intervals.

Appendix D Retrospective prediction analyses

Table 17 distinguishes row-level correctness from correctness of one causally selected answer per closed barrier. The endpoints and eligible populations differ, so the scores do not form a common architecture ranking.

Table 17. Additional retrospective prediction analyses.

Analysis	Units and tasks	Saved OOF AUC	Endpoint
Strict tabular	144,440 rows; 2,948 tasks	0.849510	Current-candidate correctness
Strict text	144,440 rows; 2,948 tasks	0.808976	Current-candidate correctness
Anonymous peer features	144,440 rows; 2,948 tasks	0.954664	Current-candidate correctness
Matched anonymous baseline	Same rows and tasks	0.945336	Current-candidate correctness
Fixed thirteen-member peers	98,280 rows; 1,512 tasks	0.940009	Current-candidate correctness
Matched fixed-roster baseline	Same rows and tasks	0.931266	Current-candidate correctness
Selected answer without timing	14,740 decisions; 2,948 tasks	0.934350	Selected-candidate correctness
Selected causal dynamics	14,740 decisions; 2,948 tasks	0.937037	Selected-candidate correctness

The strict baseline and paired peer scores were recomputed from saved predictions, with source hashes and task-fold memberships checked. The anonymous peer lift is 0.009328, with paired task-bootstrap interval [0.008028, 0.010639]; the fixed-roster lift is

0.008743, with interval [0.006653, 0.010882]. These intervals describe the archived panels. Selected-answer source and five-fold checkpoint coverage were also checked.

Strict baseline probabilities include the original task-disjoint calibration. Peer and selected-answer scores use original uncalibrated outputs; this check applies no additional reporting calibration. The later reporting calibrators use previously computed out-of-fold scores across folds and are not fully nested, so their Brier and ECE summaries are diagnostic.

Strict-baseline and selected-answer source hashes match retained code. The legacy peer runner and feature-module hashes do not match any recovered Git version, including line-ending reconstructions, although their common base script resolves to the recorded July commit. The saved arrays and paired folds are auditable; exact reproduction of the executed peer-feature pipeline remains incomplete. This qualification covers both peer populations in Table 17.

The historical peer candidate has no assigned stopping threshold. Its full-horizon collector and small smoke ledgers establish neither calibrated prospective fleet stopping nor avoided peer generation. The electronic supplement records the development history and distinguishes reused predictions from independently generated trajectories.

Appendix E Information flow and delayed repair diagrams

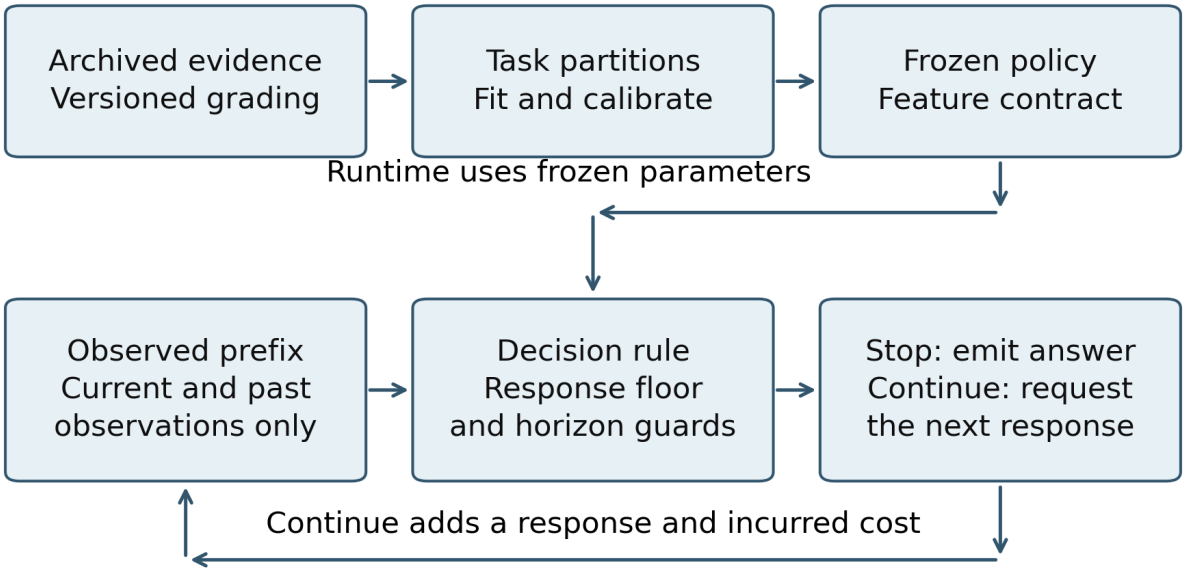


Figure 5. Offline fitting and runtime information flow. Reference labels enter fitting and evaluation offline; runtime decisions receive observed prefixes and frozen parameters. The continuation loop incurs additional generation cost. This schematic describes the information contract and does not assert a Bellman-optimal fitted controller or an executed peer fleet.

The fitted single-model controller estimates current and next selected-answer correctness and uses a one-step drift rule, with a declared response floor and horizon. The general theorem instead uses conditional multi-step continuation value.

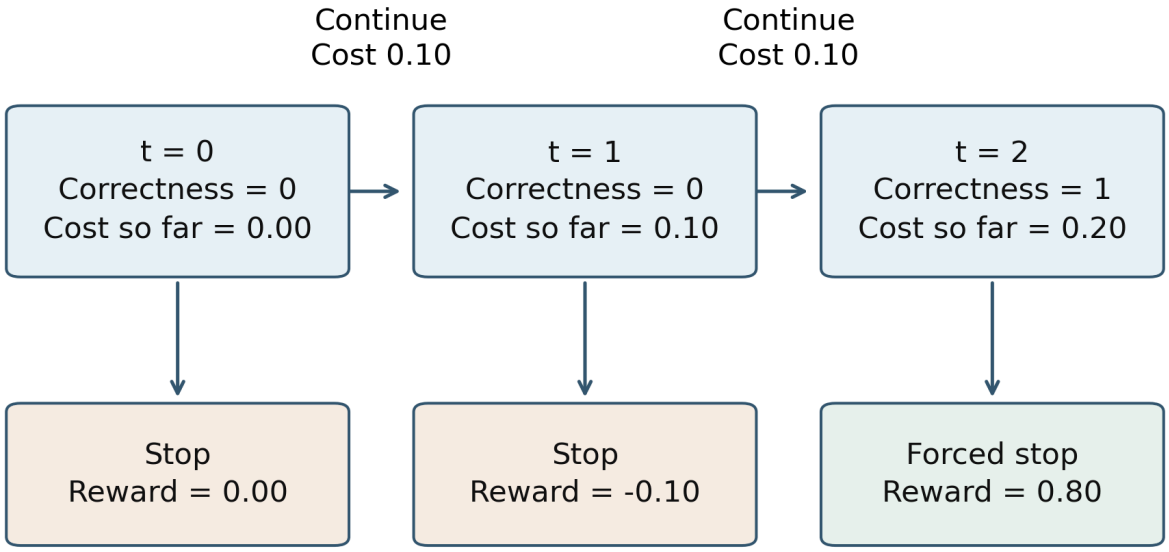


Figure 6. Delayed repair defeats a myopic stopping rule. The exact counterexample has horizon two, earliest permitted stop zero and incremental cost 0.10. Correctness follows zero, zero, one. Immediate drift at zero is -0.10, whereas continuation to the horizon yields reward 0.80. This abstract floor differs from the live experiments' floor of two; the diagram illustrates the Chapter 2 counterexample rather than new generated data.